MEMXA 5 mg/ 0.5 mL Oral Solution

ANTI-DEMENTIA

Each 0.5 mL contains:
Memantine hydrochloride 5 mg eq. to Memantine 4.16 mg

- **PRODUCT DESCRIPTION:**
Colorless to light yellowish solution

- **MECHANISM OF ACTION:**
Pharmacotherapeutic group: Other Anti-dementia drugs, ATC code: N06DX01
Pharmacology
There is increasing evidence that malfunctioning of glutamatergic neurotransmission, in particular at NMDA-receptors, contributes to both expression of symptoms and disease progression in neurodegenerative dementia.
Memantine is a voltage-dependent, moderate-affinity uncompetitive NMDA-receptor antagonist. It modulates the effects of pathologically elevated tonic levels of glutamate that may lead to neuronal dysfunction.
Pharmacokinetics
<u>Absorption</u>
Memantine has an absolute bioavailability of approximately 100 %. T_{max} is between 3 and 8 hours. There is no indication that food influences the absorption of Memantine.
<u>Distribution</u>
Daily doses of 20 mg lead to steady-state plasma concentrations of Memantine ranging from 70-150 ng/mL (0.5-1 μmol) with large interindividual variations. When daily doses of 5-30 mg were administered, a mean cerebrospinal fluid (CSF)/serum ratio of 0.52 was calculated. The volume of distribution is around 10 L/kg. About 45% of Memantine is bound to plasma-proteins.
<u>Biotransformation</u>
In man, about 80% of the circulating Memantine-related material is present as the parent compound. Main human metabolites are N-3,5-dimethyl-gludantan, the isomeric mixture of 4- and 6-hydroxy-memantine, and 1-nitroso-3,5-dimethyl-adamantane. None of these metabolites exhibit NMDA-antagonistic activity. No cytochrome P 450 catalyzed metabolism has been detected *in vitro.*
Orally administered ^{14}C-memantine, a mean of 84% of the dose was recovered within 20 days, more than 99% being excreted renally.
<u>Elimination</u>
Memantine is eliminated in a monoexponential manner with a terminal $t_{½}$ of 60-100 hours. In volunteers with normal kidney function, total clearance (Cl_{tot}) amounts to 170 mL/min/1.73 m^2 and part of total renal clearance is achieved by tubular secretion.
Renal handling also involves tubular reabsorption, probably mediated by cation transport proteins. The renal elimination rate of Memantine under alkaline urine conditions may be reduced by a factor of 7 to 9. Alkalization of urine may result from drastic changes in diet, e.g. from a carnivore to a vegetarian diet, or from the massive ingestion of alkalizing gastric buffers.
<u>Linearity</u>
It demonstrated linear pharmacokinetics in the dose range of 10-40 mg.
<u>Pharmacokinetic/pharmacodynamic relationship</u>
At a dose of Memantine of 20 mg per day the CSF levels match the k_i-value (k_i = inhibition constant) of Memantine, which is 0.5 μmol in human frontal cortex.

- **INDICATION:**
Treatment of patients with moderate to severe Alzheimer's disease.

- **DOSAGE AND ADMINISTRATION:**
Oral
Treatment should be initiated and supervised by a physician experienced in the diagnosis and treatment of Alzheimer's dementia.
<u>Posology</u>
Therapy should only be started if a caregiver is available who will regularly monitor the intake of the medicinal product by the patient. Diagnosis should be made according to current guidelines. The tolerance and dosing of Memantine should be reassessed on a regular basis, preferably within three months after start of treatment. Thereafter, the clinical benefit of Memantine and the patient's tolerance of treatment should be reassessed on a regular basis according to current clinical guidelines. Maintenance treatment can be continued for as long as a therapeutic benefit is favorable and the patient tolerates treatment with Memantine. Discontinuation of Memantine should be considered when evidence of a therapeutic effect is no longer present or if the patient does not tolerate treatment.
Adults
<u>Dose titration</u>
The maximum daily dose is 20 mg per day. In order to reduce the risk of undesirable effects the maintenance dose is achieved by upward titration of 5 mg per week over the first 3 weeks as follows:
Week 1 (day 1-7)
The patient should take 0.5 mL solution (5 mg) per day for 7 days.
Week 2 (day 8-14)
The patient should take 1 mL solution (10 mg) per day for 7 days.
Week 3 (day 15-21)
The patient should take 1.5 mL solution (15 mg) per day for 7 days.
From Week 4 on
The patient should take 2 mL solution (20 mg) once a day.
<u>Maintenance dose</u>
The recommended maintenance dose is 20 mg per day.
Elderly
The recommended dose for patients over the age of 65 years is 20 mg per day (2 mL solution) as described above.
Pediatric population
The safety and efficacy of Memantine oral solution children below 18 years old has not yet been established.
Renal impairment
In patients with mildly impaired renal function (creatinine clearance 50-80 mL/min) no dose adjustment is required. In patients with moderate renal impairment (creatinine clearance 30-49 mL/min) daily dose should be 10 mg (1 mL solution). If tolerated well after at least 7 days of treatment, the dose could be increased up to 20 mg/day according to standard titration scheme. In patients with severe renal impairment (creatinine clearance 5-29 mL/min) daily dose should be 10 mg (1 mL solution) per day.
Hepatic impairment
In patients with mild or moderate hepatic impaired function (Child-Pugh A and Child-Pugh B) no dose adjustment is needed. No data on the use of Memantine in patients with severe hepatic impairment are available. Administration of Memantine is not recommended in patients with severe hepatic impairment.
<u>Method of administration</u>
Memantine should be taken orally once daily at the same time each day. The solution can be taken with or without food. The solution directly from the bottle or syringe poured where applicable.
Follow the detailed instructions on the preparation and handling of the product.

Fig. 1: Prior to first use, remove the screw cap from the bottle; the cap must be turned anticlockwise and unscrewed completely.

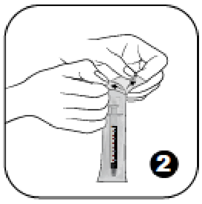

Fig. 2: Remove the 1 mL syringe from the plastic bag.

Fig. 3: Using the 1 mL syringe, pull out the amount prescribed in the dose titration.

Fig. 4 and 5. A glass with a little water or a spoon should be used to dispense the desired amount of Memantine hydrochloride solution (0.5 mL of solution equivalent to Memantine hydrochloride 5 mg) to the patient.

♦ WARNING AND PRECAUTION:

Caution is recommended in patients with epilepsy, former history of convulsions or patients with predisposing factors for epilepsy.

Concomitant use of N-methyl-D-aspartate (NMDA)-antagonists such as Amantadine, Ketamine or Dextromethorphan should be avoided. These compounds act at the same receptor system as Memantine, and therefore adverse reactions (mainly central nervous system (CNS)-related) may be more frequent or more pronounced.

Some factors that may raise urine pH may necessitate careful monitoring of the patient. These factors include drastic changes in diet, e.g. from a carnivore to a vegetarian diet, or a massive ingestion of alkalizing gastric buffers. Also, urine pH may be elevated by states of renal tubulary acidosis (RTA) or severe infections of the urinary tract with *Proteus* bacteria.

Solution contains Sorbitol and Potassium

The medicine contains Sorbitol. Patients with rare hereditary problems of fructose intolerance should not take this medicine.

Furthermore, this medicine contains Potassium, less than 1 mmol per dose i.e. essentially potassium-free.

♦ PREGNANCY AND LACTATION:

Pregnancy

The potential risk for humans is unknown. Memantine should not be used during pregnancy unless clearly necessary.

Breastfeeding

It is not known whether Memantine is excreted in human breast milk but, taking into consideration the lipophilicity of the substance, this probably occurs. Women taking Memantine should not breastfeed.

♦ SIDE EFFECT:

Adverse reactions are ranked according to system organ class, using the following convention: very common (≥ 1/10), common (≥ 1/100 to < 1/10), uncommon (≥ 1/1,000 to < 1/100), rare (≥ 1/10,000 to < 1/1,000), very rare (< 1/10,000), no known (cannot be estimated from the available data).

SYSTEM ORGAN CLASS	FREQUENCY	ADVERSE REACTION
Infections and infestations	Uncommon	Fungal infections
Immune system disorders	Common	Drug hypersensitivity
Psychiatric disorders	Common	Somnolence
	Uncommon	Confusion
	Uncommon	Hallucinations[1]
	Not known	Psychotic reactions[2]
Nervous system disorders	Common	Dizziness
	Common	Balance disorders
	Uncommon	Gait abnormal
	Very rare	Seizures
Cardiac disorders	Uncommon	Cardiac failure
Vascular disorders	Common	Hypertension
	Uncommon	Venous thrombosis/thromboembolism
Respiratory, thoracic and mediastinal disorders	Common	Dyspnea
Gastrointestinal disorders	Common	Constipation
	Uncommon	Vomiting
	Not known	Pancreatitis[2]
Hepatobiliary disorders	Common	Elevated liver function test
	Not known	Hepatitis
General disorders and administration site conditions	Common	Headache
	Uncommon	Fatigue

[1] Hallucinations have mainly been observed in patients with severe Alzheimer's disease.

[2] Isolated cases reported in post-marketing experience.

Alzheimer’s disease has been associated with depression, suicidal ideation and suicide. In post-marketing experience these reactions have been reported in patients treated with Memantine.

♦ EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Moderate to severe Alzheimer’s disease usually causes impairment of driving performance and compromises the ability to use machinery. Furthermore, Memantine has minor or moderate influence on the ability to drive and use machines such that outpatients should be warned to take special care.

- **CONTRAINDICATION:**

Hypersensitivity to the active substance or to any of the excipients.

- **DRUG INTERACTION:**

Due to the pharmacological effects and the mechanism of action of Memantine the following interactions may occur:

- The mode of action suggests that the effects of L-dopa, dopaminergic agonists, and anticholinergics may be enhanced by concomitant treatment with NMDA-antagonists such as Memantine. The effects of barbiturates and neuroleptics may be reduced. Concomitant administration of Memantine with the antispasmodic agents, Dantrolene or Baclofen, can modify their effects and a dose adjustment may be necessary.
- Concomitant use of Memantine and Amantadine should be avoided, owing to the risk of pharmacotoxic psychosis. Both compounds are chemically related NMDA-antagonists. The same may be true for Ketamine and Dextromethorphan. There is one published case report on a possible risk also for the combination of Memantine and Phenytoin.
- Other active substances such as Cimetidine, Ranitidine, Procainamide, Quinidine, Quinine and nicotine that use the same renal cationic transport system as Amantadine may also possibly interact with Memantine leading to a potential risk of increased plasma levels.
- There may be a possibility of reduced serum level of Hydrochlorothiazide (HCT) when Memantine is co-administered with HCT or any combination with HCT.
- Isolated cases with international normalized ratio (INR) increases have been observed in patients concomitantly treated with Warfarin. Although no casual relationship has been established, close monitoring of prothrombin time or INR is advisable for patients concomitantly treated with oral anticoagulants.
- No relevant active substance-active substance interaction of Memantine with Glyburide/Metformin or Donepezil was observed.
- No relevant effect of Memantine on the pharmacokinetics of Galantamine was observed.
- Memantine did not inhibit CYP 1A2, 2A6, 2C9, 2D6, 2E1, 3A, flavin containing monooxygenase, epoxide hydrolase or sulphation *in vitro*.

- **OVERDOSE AND TREATMENT:**

Symptoms

Relative large overdoses (200 mg and 105 mg/day for 3 days, respectively) have been associated with either only symptoms of tiredness, weakness and/or diarrhea or no symptoms. In the overdose cases below 140 mg or unknown dose the patients revealed symptoms from central nervous system (confusion, drowsiness, somnolence, vertigo, agitation, aggression, hallucination, and gait disturbance) and/or of gastrointestinal origin (vomiting and diarrhea).

In the most extreme case of overdose, the patient survived the oral intake of a total of 2,000 mg Memantine with effects on the central nervous system (coma for 10 days, and later diplopia and agitation). The patient received symptomatic treatment and plasmapheresis. The patient recovered without permanent sequelae.

In another case of a large overdose, the patient also survived and recovered. The patient had received 400 mg Memantine orally. The patient experienced central nervous system symptoms such as restlessness, psychosis, visual hallucinations, proconvulsiveness, somnolence, stupor, and unconsciousness.

Treatment

In the event of overdose, treatment should be symptomatic. No specific antidote for intoxication or overdose is available. Standard clinical procedures to remove active substance material, e.g. gastric lavage, carbo medicinalis (interruption of potential entero-hepatic recirculation), acidification of urine, forced diuresis should be used as appropriate.

In case of signs and symptoms of general central nervous system (CNS) overstimulation, careful symptomatic clinical treatment should be considered.

- **STORAGE:**

Store at temperature of not more than 30°C.

Once opened, the contents of the bottle should be used within 3 months.

- **DOSAGE FORM AND PACKAGING AVAILABLE:**

Bottle by 50 mL (with 1.0 mL Syringe)

- **DATE OF REVISION**

June 26, 2023

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand

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